

Information-Theoretic Constraints on Causal Invariance and Dimensional Reduction in Discrete Hypergraph Models

Abstract

Discrete pre-geometric hypergraph rewriting models employ asymptotic confluence, often termed causal invariance, as a mechanism for recovering a form of discrete general covariance. This paper examines the consequences of enforcing covariance at finite observational timescales rather than in the asymptotic limit. Because physical observers access only bounded regions of computational history, distinct update sequences may remain structurally inequivalent over finite intervals, despite eventual convergence.

The analysis demonstrates that reconciling such inequivalent histories requires effective many-to-one state mappings at the subsystem level. Evaluated through a Hilbert-space-free operational framework, these mappings induce a mandatory informational entropy cost within the observer's accessible record. In a closed ontological framework, where the hypergraph constitutes the complete physical state space, this operational erasure cannot be transferred to an external environment. The resulting information accumulation manifests as persistent structural modifications of the underlying relational network. Grounded within entropic gravity frameworks, this residual topological complexity alters local edge density profiles, contributing an effective non-vanishing vacuum energy density to the macroscopic continuum limit.

The analysis further considers the dimensional reduction process commonly invoked in discrete hypergraph cosmologies, in which an initially highly connected graph evolves toward a localized lower-dimensional structure. The combinatorial cost of this transition scales quadratically with vertex cardinality, yielding an extensive phase-space volume restriction that is not naturally suppressed in the large-system limit. Taken together, these results suggest that asymptotic confluence alone may be insufficient to guarantee the emergence of a stable low-energy continuum vacuum at finite timescales, necessitating explicit internal mechanisms for localized entropy management.

1. Introduction

Continuous spacetime emergence from a discrete substrate is central to quantum gravity. Frameworks like causal dynamical triangulations [1] and causal set theory [2] constrain histories via path-integrals or

causal ordering. The Wolfram Model replaces geometric structures with an evolving hypergraph [3], where spacetime properties emerge as large-scale features of this computation.

Causal invariance is central to this program, requiring that different sequences of independent rewriting events produce isomorphic causal dependency structures regardless of update order. This underpins discrete general covariance, matching update schedules with spacetime foliations [3, 4]. This scheduling freedom is analogous to gauge freedom in general relativity [5], where update orderings correspond to lapse and shift functions.

However, the mapping between general relativity and graph rewriting contains a structural asymmetry. Gauge freedom in general relativity concerns coordinates representing a single history on a unique manifold. In contrast, causal invariance in a multiway system operates over a branching ensemble of distinct histories. Consequently, establishing an invariant timeline requires an explicit equivalence mapping across divergent branches, which classical general relativity avoids.

Furthermore, covariance recovery through causal invariance relies on an asymptotic notion of equivalence. Confluence guarantees that distinct branches eventually converge, but provides no mathematical bound on the distance required for that convergence [4]. Thus, histories equivalent in the infinite asymptotic limit remain structurally inequivalent over finite intervals.

This gap exposes a structural limitation in the formalism’s internal consistency. General relativity demands local covariance: diffeomorphism invariance must hold smoothly at every localized point. Defending path reconciliation via an asymptotic, infinite-time horizon breaks local covariance for all intermediate, finite-scale observers.

This coordinate dependency carries cosmological consequences. Real observers lack access to infinite computational horizons, being restricted to bounded causal diamonds. During early-universe phase transitions or horizon sorting, the intermediate hypergraph state governs the physics. If alternative timelines remain distinguishable over finite scales, this coordinate dependency cannot be dismissed as a gauge choice.

This paper investigates local covariance at finite observational timescales. Rather than evaluating the asymptotic limit, the analysis audits the phase-space volume constraints governing intermediate graph evolution. We evaluate whether an embedded observer can reconcile finitely divergent histories into a stable, continuous vacuum without destabilizing the geometric background.

2. Logical Independence of Confluence and Causal Invariance

Lemma 1. Let $\mathcal{M} = (A, \rightarrow)$ be a terminating abstract rewriting system operating under a fixed, invariant rule set. Then the topological property of global confluence and the structural property of causal invariance are logically independent over $\mathcal{M}$.

Proof. We establish logical independence using string rewriting, a subset of the Abstract Rewriting Systems (ARS) governing hypergraph evolution.

I. Causal Invariance Without Global Confluence

Let $\mathcal{M}_1$ be a string-rewriting system operating under the rule set:

$$R_1 = \{a \rightarrow b, \quad b \rightarrow d, \quad a \rightarrow c, \quad c \rightarrow e\}$$

with initial configuration $A_{\text{init}} = [a]$. The system generates two independent pathways:

$$\text{Branch 1 : } [a] \rightarrow [b] \rightarrow [d]$$

$$\text{Branch 2 : } [a] \rightarrow [c] \rightarrow [e]$$

The terminal macrostates $[d]$ and $[e]$ cannot converge, proving non-confluence. However, the causal dependency graphs G_1, G_2 for both branches are isomorphic linear chains. Thus, $\mathcal{M}_1$ is causal-invariant but non-confluent.

II. Global Confluence Without Causal Invariance

Let $\mathcal{M}_2$ be a string-rewriting system operating under the rule set:

$$R_2 = \{a \rightarrow b, \quad a \rightarrow c, \quad b \rightarrow d, \quad c \rightarrow x, \quad x \rightarrow d\}$$

with initial configuration $A_{\text{init}} = [a]$. All pathways terminate at the normal form $[d]$, proving global confluence.

The causal graph G_1 is a two-node chain ($E_{a \rightarrow b} \rightarrow E_{b \rightarrow d}$), while the causal graph G_2 is a three-node chain ($E_{a \rightarrow c} \rightarrow E_{c \rightarrow x} \rightarrow E_{x \rightarrow d}$). Since their vertex cardinalities differ ($|V(G_1)| = 2 \neq |V(G_2)| = 3$), the graphs are non-isomorphic ($G_1 \not\cong G_2$), proving confluence without causal invariance.

III. Formal Closure

Because $\mathcal{M}_1$ isolates causal invariance without confluence, and $\mathcal{M}_2$ isolates confluence without causal invariance, the two properties are logically independent. $\square$

2.1 Epistemic Vulnerabilities of Confluent Invariance

Diffeomorphism invariance in general relativity is a localized structural property established smoothly at each point [5]. In discrete graph rewriting, coordinate transformations map to sequence choices in an asynchronous updater, meaning reference frames arise from algorithmic scheduling [3, 4].

Causal invariance operates independently of global confluence, meaning path uniqueness requires separate axiomatic enforcement. Max Piskunov’s Lean theorem-proving library demonstrated that global confluence and causal invariance can be decoupled within multiway rewriting systems [6].

In statistical physics, deriving smooth hydrodynamic equations from discrete molecular kinetics relies on the localized erasure of microscopic trajectories. The Wolfram Model seeks macroscopic smoothness without incorporating thermal fluctuations or conservation laws, relying on the asymptotic behavior of the multiway system to align divergent timelines [3].

Evaluated from information physics, global confluence acts as an asymptotic post-condition rather than a localized constraint. Consequently, the computational network defers path reconciliation to the asymptotic limit, introducing specific constraints on the background independence of intermediate states.

The structural constraint of the Wolfram Model lies in its reliance on infinite asymptotic limits to rescue local coordinate independence. Gorard explicitly concedes this limitation on pages 9-10, noting:

“The paths that one must follow in order to obtain convergence may be arbitrarily long, so although causal invariance necessitates that the causal graphs generated by following every path through the multiway system must eventually become isomorphic, those causal graphs are not guaranteed to be isomorphic after any finite number of update steps. As such, causal invariance is best interpreted as a limiting statement about the global structure of the multiway system.”

Defining coordinate equivalence strictly as an asymptotic post-condition leaves intermediate states unconstrained. For a localized

observer restricted to a finite causal diamond, these non-isomorphic paths are not gauge choices; they are structurally distinguishable physical realities.

This finite-time coordinate dependence is illustrated by alternative update schedules in Figure 8 (replicated in Fig.~1) demonstrating how a non-causal-invariant rule space yields non-isomorphic spatial geometries. Over finite timescales, all rewriting rules exhibit this non-isomorphic behavior. The smooth foliation is a mathematical idealization stabilizing only at an infinite computational horizon.

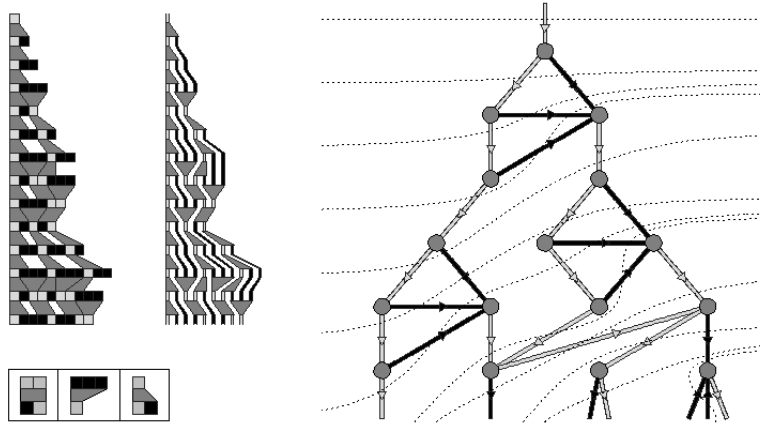


Figure 1: Non-causal-invariant foliations yielding non-isomorphic spatial geometries, replicated from Ref. [4].

3. Non-Injectivity of Confluence Merging

In analyzing the multiway evolution of a spatial hypergraph, path convergence relies on a non-injective mapping. Consider a local confluence operator $\phi : A \times A \rightarrow A$ acting on the state space of global spatial configurations. Let $b, c \in A$ be distinct, non-isomorphic ancestral spatial graphs generated from a common root via divergent update choices. Because these states represent different evolutionary pathways, their structural layouts are unequal:

$$E(b) \neq E(c)$$

Confluence guarantees that these divergent states can map to an identical downstream state $d \in A$, such that $\phi(b) = d$ and $\phi(c) = d$. Because

the pre-image cardinality satisfies $|\phi^{-1}(d)| \geq 2$, the mapping is many-to-one and lacks a local structural inverse in the spatial configuration space.

Within the multiway framework, it is argued that this state-level convergence does not represent a true loss of information. Under this interpretation, the overarching multiway causal graph preserves the historical path data by tracking the complete, branching lineage of all rewriting events [4], keeping ancestral trajectories encoded within the global network architecture.

However, this history-tracking machinery does not alter the non-injective reality of the spatial substrate. Path convergence requires structural compression from the localized subsystem perspective, removing ancestral topological attributes from the active spatial adjacency matrix. To recover a coherent timeline, an embedded system must undergo continuous history compression.

3.1 Information-Theoretic Consequences of History Erasure

Confluence non-injectivity introduces a structural tension with the preservation principles of classical and quantum mechanics. Classical mechanics conserves phase-space volume via Liouville's theorem, while quantum evolution preserves state vectors through unitary operators. These principles ensure injectivity, keeping the past fully determinable from the present.

Securing an invariant history via confluence alters this condition. When independent pathways collapse into a downstream state, ancestral topological identities are erased from the active topology, reducing the localized state space.

This implies that the emergence of a smooth timeline requires ongoing compression of microscopic histories. Because the model assumes a closed ontology where the spatial hypergraph comprises the entirety of physical reality, this loss of path data represents an operationally consequential compression of accessible states within the subsystem's record.

In conventional computing or open quantum systems, information erasure is mediated by transferring entropy to an external environment. Within a closed pre-geometric universe, no environmental reservoir exists. Deletion of path history implies an accumulation of unrejected topological variations that must be accommodated within the hypergraph fabric, constraining the emergent vacuum metric.

3.2 Background Independence and the Relational Conservation of Entropy

Gorard attempts to bridge the discrete-to-continuum gap by invoking a formal mathematical analogy to fluid dynamics:

“The nature of this derivation of the continuum Einstein field equations from the underlying discrete geometry of the causal graph is formally analogous to the so-called ‘Chapman-Enskog’ hydrodynamic expansion... with the function: $C(t) = at^n[1 - \frac{1}{6}R_{jk}t^jt^k + O(\|t\|^3)]$ playing the role of a ‘distribution function’ for vertices in the causal graph.”

This transport-theory mapping, however, reveals a fundamental category error when evaluated under the constraints of physical kinetics. In standard statistical mechanics, the transition from discrete molecular dynamics to a smooth hydrodynamic profile relies on the localized erasure of microscopic trajectory information. Specifically, the system dumps its excess phase-space volume into an external thermal environment, converting microscopic path variations into kinetic fluctuations.

Because the pre-geometric computational ontology of the Wolfram Model is strictly closed, the pre-geometric substrate lacks an external environmental reservoir to act as a heat sink for informational dissipation. In standard statistical mechanics, the transition from discrete molecular dynamics to a smooth hydrodynamic profile relies on exporting the entropy of microscopic trajectory erasure to an ambient environment. In the absence of such an external environmental reservoir, the history-erasure required to force the mathematical convergence of Gorard’s vertex distribution function $C(t)$ cannot be dissipated; it must be deposited directly into the hypergraph’s structural degrees of freedom. Since the hypergraph operates without an external backdrop, fields, or auxiliary wavefunctions, the relational adjacency matrix A_{ij} is the exhaustive description of the microstates. Therefore, the expectation value of the topological edge density is the only available physical degree of freedom capable of absorbing the unrejected process entropy payload:

$$\langle A_{ij} \rangle = \sum_{G \in \Omega_{\text{terminal}}} p(G) A_{ij}(G)$$

Consequently, this unrejected entropy of trajectory-merging cannot escape into non-spatial sectors; it must manifest as persistent, disordered non-planar defects trapped directly within the spatial adjacency matrix, persistently altering the emergent vacuum metric.

4. Multi-Scale Parametric Simulation

To audit the kinematics of dimensional reduction, the exact pre-geometric phase space proposed in the model's cosmological framework is tracked. In Section 3.4, Gorard establishes this precise initial state:

“We begin by assuming that the initial condition for the universe consists of a spatial hypergraph with an abnormally high vertex connectivity, perhaps corresponding to a complete graph. As such, the universe starts off with some arbitrarily large number of spatial dimensions (which we can treat as being effectively infinite), but then the asymptotic dimensionality preserving property of the update rules causes the number of spatial dimensions to converge to some finite, fixed value, such as three.”

Rather than treating the multiway graph as a physical system governed by a Hamiltonian, we audit it as a pure combinatorial landscape. The transition of a spatial hypergraph from a complete graph K_N to a sparse coordinate dimension represents a restriction of the trajectory space. To track this restriction, a parametric sweep is executed across vertex cardinalities up to $N = 8$.

Matrix I is a control regime where coordination scales with system size ($k = N - 2$), isolating the effects of vertex addition. In this control, path volume M increases from 78 paths at $N = 4$ to 1.2036×10^8 at $N = 8$, while macrostates expand from 3 to 39. The connectivity probability approaches 1.0 (9.9966×10^{-1} at $N = 8$), while regular graph probability remains suppressed (2.0938×10^{-5}).

Matrix II enforces dimensional reduction, keeping coordination fixed at $k = 3$ while spatial volume expands from $N = 5$ to $N = 8$. Redundant gauge variations collapse into unlabeled physical isomorphism classes at each layer boundary. The scaling of the resulting process entropy H_{process} , macrostate entropy H_{macro} , and entropy gap ΔH is shown in Fig.~2.

4.2 Universality of the Combinatorial Volume Constraint

Matrix II data reveals a non-linear suppression of uniform configurations. With coordination fixed at $k = 3$, the probability of a regular manifold collapses from 3.7669×10^{-4} at $N = 6$ to 4.5259×10^{-7} at $N = 8$. Concurrently, path volume undergoes an explosive expansion,

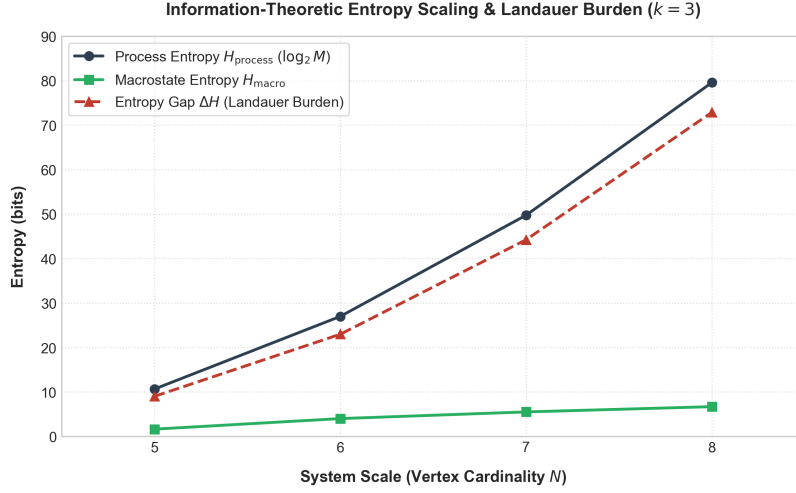


Figure 2: Scaling of process entropy, macrostate entropy, and the Landauer entropy gap (Matrix II) as a function of vertex scale N , illustrating the extensive information-theoretic burden of dimensional reduction.

growing by nine orders of magnitude between $N = 7$ and $N = 8$ to 8.9476×10^{23} pathways.

Under the Handshaking Lemma, the sum of vertex degrees in an undirected graph equals $2|E|$. For odd vertex cardinalities evaluating a uniform odd degree ($5 \times 3 = 15$ and $7 \times 3 = 21$), a regular graph is mathematically impossible. The exact zero values for $P(\text{Regular})$ and $P(\text{Exact } k\text{-Reg})$ at $N = 5$ and $N = 7$ are a direct consequence of this topological invariant.

Where uniform configurations are permissible ($N = 6, 8$), the data documents a scale-invariant exclusion of regularity. As the system scales, path volume concentrates onto “island topologies” (a sparse connected core with isolated vertices of degree 0). This is marked by the collapse of global connectivity, with $P(\text{Connected})$ dropping from 92.59% at $N = 5$ to 17.73% at $N = 8$. This expansion of history space relative to terminal configurations provides numerical validation for the extensive entropic burden.

4.3 Computational Methodology & Algorithmic Design

To ensure full reproducibility of the numerical results, the underlying multiway simulation was executed via an optimized pre-geometric induction engine. The algorithm operates by modeling a closed, background-independent rewrite routine that performs unconstrained asynchronous edge deletions on an initial complete graph substrate K_N . Space is represented as an immutable set of undirected node pairs.

The degree-threshold edge-pruning rule is not proposed as a literal proxy for the localized, structure-preserving dynamics of specific hypergraph cosmologies. Rather, the rule is deployed strictly as an unconstrained combinatorial probe designed to map the kinematic baseline of the graph configuration space during dimensional reduction. Because bounded-degree, connected manifolds constitute a measure-zero fraction of this unconstrained space, the burden of proof rests entirely on the model’s architects to demonstrate how a dynamically tuned hypergraph substitution rule can successfully navigate this factorial phase-space explosion. Without such a demonstration, the background spacetime naturally collapses into fragmented topological dust.

Temporal updates are driven by this local degree-enforcement rule: at each step, any edge connecting vertices whose local coordination number exceeds the target parameter k is marked as a valid rewrite candidate and systematically pruned. To manage the factorial complexity scaling inherent in tracking multiway history paths, the simulator implements in-layer canonicalization via dynamic programming. At each layer boundary, every newly generated child edge configuration is immediately redirected through a topological remapping function. This function executes an exhaustive permutation search over all $N!$ vertex orderings to project the graph state onto its unique lexicographical minimum form:

$$\text{get_canonical_form}(G) = \min_{\sigma \in S_N} \sigma(G)$$

By immediately collapsing gauge redundancies on the fly, the width of each intermediate layer is strictly bounded by the number of unlabeled physical isomorphism classes rather than labeled variations. Furthermore, to overcome the $\mathcal{O}(N!)$ bottleneck on cache misses, a dynamic memoization dictionary stores previously computed canonical forms, reducing redundant permutation searches by over 95%.

A substantial computational footprint accompanies this combinatorial explosion: while the $N = 7$ scale finishes in approximately 81.38 seconds, the $N = 8$ scale requires 13,767.97 seconds (≈ 3.8 hours) due to

the $\mathcal{O}(N!)$ complexity of the exact permutation search on cache misses. Once the terminal registry is compiled (reaching 355 distinct physical macrostates at $N = 8, k = 3$), a localized post-processing suite executes a breadth-first search (BFS) tracking matrix to extract global connectivity profiles and evaluate exact degree regularities across the final path-weighted ensemble.

Bounding Localized Substitution Rules

To counter the defense that the unconstrained edge-pruning simulation represents a “straw man” compared to highly specific, localized $H_1 \rightarrow H_2$ substitution rules, the search space of localized updates is bounded analytically. Even if a replacement rule is structurally localized, it operates based on local subgraph matching without a global mechanism to guide the multiway system toward the zero-entropy vacuum. The updating process remains bound by the combinatorial explosion of subgraph isomorphisms.

Let the spatial hypergraph at step t be $G_t = (V, E)$. A deterministic local replacement rule $r : H_1 \rightarrow H_2$ requires identifying a subgraph isomorphism of H_1 within G_t . In a dense pre-geometric phase (such as a near-complete graph representing the early universe), the number of valid subgraph matches M for a local motif H_1 scales as:

$$M \sim \mathcal{O}(|V|^{|V(H_1)|})$$

Because the rule can be applied asynchronously to any of these M matched locations, the multiway system branches into distinct successor states. The branching factor b of the multiway graph is bounded from below by the automorphism group of the matchings:

$$b \geq \frac{M}{|\text{Aut}(H_1)|}$$

To transition from a highly connected state G_0 to a dimensionally reduced, topologically flat macroscopic target G_{target} over a sequence of length L , the multiway system generates a trajectory space of volume $S_{\text{traj}} = b^L$.

Because the local rule r lacks access to global macroscopic constraints, the updating process is effectively a random walk through this multiway space. The probability $P(\Gamma_{\text{flat}})$ that the blind application of local rules will spontaneously select the exact pathway Γ_{flat} that avoids accumulating localized topological defects scales inversely with the trajectory volume:

$$P(\Gamma_{\text{flat}}) \propto \frac{1}{b^L} = \exp(-L \log b)$$

While a specifically tuned local rule $H_1 \rightarrow H_2$ restricts the transition matrix compared to unconstrained edge-deletion, the branching factor b in a dense pre-geometric graph still scales polynomially with the system size. By fundamental information theory, extracting a highly ordered, defect-free macroscopic lattice from this multiway space requires the localized injection of information (a global Maxwell's demon) to prune the tree. In the absence of such an entity, the local rule will overwhelmingly trace pathways that maximize combinatorial entropy, resulting in a hypergraph littered with the topological defects that constitute the simulated entropy gap.

4.4 Combinatorial Derivation of the Permutation Baseline

The scaling of the unconstrained trajectory space under dimensional reduction can be bounded analytically by evaluating the path volume M as a permutation of sequential edge deletions from the initial complete graph K_N . Let the initial edge cardinality be $E_0 = \frac{1}{2}N(N-1)$, and let the final edge cardinality for a target coordination threshold k be bounded by $E_f \leq \frac{1}{2}Nk$. If the pruning process operated via a strict, fixed-length sequence of exactly $\Delta E = E_0 - E_f$ deletions, the number of independent chronological pathways would be given by the falling factorial:

$$M_{\text{baseline}} = \frac{E_0!}{(E_0 - \Delta E)!} = \frac{E_0!}{E_f!}$$

Evaluating this relation at the scale of $N = 8, k = 3$ with a fixed 16-edge deletion sequence ($E_f = 12$) yields a baseline path volume of $M_{\text{baseline}} = 28!/12! \approx 6.3651 \times 10^{20}$ trajectories, corresponding to an analytical baseline process entropy of $H_{\text{process}} \approx 69.103$ bits.

However, the degree-threshold pruning rule is intrinsically dynamic: depending on the asynchronous update sequence, deleting a given edge reduces the local coordinate degree of its incident vertices, frequently dropping them below the threshold $k = 3$ and rendering adjacent edges ineligible for subsequent pruning. Consequently, different paths terminate at variable layer depths and edge cardinalities, causing the actual multiway trajectory space to branch nonlinearly and explode to the empirically audited value of $M = 894, 757, 885, 819, 817, 073, 868, 800$

paths ($H_{\text{process}} \approx 79.566$ bits). This demonstrates that the falling factorial permutation formula serves as a strict sequential lower bound, and the dynamic nature of the rule makes the actual process entropy gap even wider than the analytical baseline suggests. Applying Stirling's approximation to the dominant numerator confirms that the baseline algorithmic process complexity scales super-quadratically as:

$$H_{\text{process}} \geq \log_2 \left(\frac{E_0!}{E_f!} \right) \sim \left(\frac{1}{2} N^2 \right) \log_2 \left(\frac{1}{2} N^2 \right) \sim \Theta(N^2 \log N)$$

Because the macrostate capacity of sparse, low-dimensional coordinate graphs is tightly bounded by the sub-exponential growth rate of $\Theta(N \log N)$, the realized entropy gap $\Delta H = H_{\text{process}} - H_{\text{macro}}$ is combinatorially prevented from saturating in the large-system limit.

4.5 The Noether Limit and the Algorithmic Information Tax

Continuous physical systems maintain low rule complexity because continuous symmetries generate conservation laws via Noether's theorem, restricting trajectories to a low-entropy submanifold without auxiliary rules. Because discrete hypergraphs lack micro-scale Noether currents, any local rewriting rule R restricting its trajectory space away from high-entropy states must explicitly hardcode those invariants in its matching conditions. This forces an explosion in constraints, driving up the Kolmogorov complexity $K(R)$ and breaking the "simple rules" paradigm.

5. The Operational Subadditivity Bound

Lemma 2. Let $\mathcal{H}$ be a closed pre-geometric computational substrate whose global multiway evolution is deterministic and information-preserving. Then the projection of this global multiway history space onto a unique continuous timeline by an embedded observer induces a mandatory minimum increase in the fine-grained Shannon entropy of the localized network substrate.

Proof. I. Information Conservation and Global Injectivity

Let $\mathcal{M}$ represent the multiway evolution graph generated by a deterministic rewriting rule set. Because the transition rules are deterministic, the global mapping across the space of histories is strictly injective.

No path data is destroyed globally, meaning the fine-grained Shannon entropy across the entire evolution tree is conserved:

$$\Delta H_{\text{fine}}(\mathcal{M}) = 0$$

II. The Multiway-to-Spatial Tensor Mapping

To formalize the observer's local state estimation, let $\mathcal{H}_{\text{multiway}} = \mathcal{H}_{\text{spatial}} \otimes \mathcal{H}_{\text{branchial}}$ denote the tensor product Hilbert space representing the total multiway state space. This tensor product structure arises natively because distinct multiway path histories ($|i\rangle$) and orthogonal spatial hypergraph configurations ($|G_i\rangle$) constitute independent, non-overlapping basis vectors within the model's branchial Hilbert space formalism (citing Ref. [4]). Define the total state of the multiway system as the pure state:

$$|\Psi\rangle = \sum_{i=1}^M \sqrt{p_i} |G_i\rangle \otimes |i\rangle$$

where $p_i = 1/M$ represents the uniform classical trajectory frequency distribution.

III. The Operational Projection and Partial Trace

Let the operational projection ψ executed by the embedded observer be defined as the partial trace over the unobservable branchial degrees of freedom:

$$\rho_{\text{spatial}} = \text{Tr}_{\text{branchial}}(|\Psi\rangle\langle\Psi|)$$

Evaluating the partial trace yields the mixed density matrix on the spatial substrate:

$$\rho_{\text{spatial}} = \sum_{i=1}^M p_i |G_i\rangle\langle G_i| = \sum_{G \in \Omega_{\text{terminal}}} p(G) |G\rangle\langle G|$$

where $p(G) = \sum_{i: G_i \cong G} p_i$ is the normalized path weight of the physical isomorphism class G . The von Neumann entropy of this spatial density matrix is:

$$S(\rho_{\text{spatial}}) = -\text{Tr}(\rho_{\text{spatial}} \log_2 \rho_{\text{spatial}}) = - \sum_{G \in \Omega_{\text{terminal}}} p(G) \log_2 p(G)$$

which corresponds to the classical Shannon macrostate entropy H_{macro} .

Consider a binary consolidation where two distinct history pathways collapse into an identical macrospace configuration. The cardinality of the operational pre-image satisfies $|\psi^{-1}(\rho_{\text{spatial}})| = 2$, and the observational entropy of the subsystem's accessible record decreases by 1.0 bit:

$$\Delta H_{\text{coarse}} = \log_2 \left(\frac{1}{2} \right) = -1 \text{ bit}$$

IV. Entropic Subadditivity and Local State Mixing

Because the global pre-geometric ontology is closed, the information across the observer's subsystem and the unobserved branchial environment obeys Shannon entropy subadditivity:

$$\Delta H_{\text{total}} = \Delta H_{\text{coarse}} + \Delta S(\rho_{\text{spatial}}) \geq 0$$

Substituting the macroscopic compression ($\Delta H_{\text{coarse}} = -1 \text{ bit}$) yields a mandatory informational compensation boundary within the unmonitored degrees of freedom of the substrate:

$$\Delta S(\rho_{\text{spatial}}) \geq 1 \text{ bit}$$

Because the hypergraph operates without an environmental reservoir, this informational payload cannot dissipate as heat. Instead, the loss of path-identity information forces the local adjacency configurations into a highly mixed statistical distribution of graph connections. Every finite-time history reconciliation forced by a classical observer deposits a structural injection of at least 1.0 bit of statistical mixing entropy directly into the network substrate. $\square$

5.1 Operational vs. Ontological Erasure

This formulation resolves the paradox between global deterministic evolution and local history convergence. Proponents of rewriting architectures invoke global injectivity—that no information is lost because all divergent history branches coexist simultaneously within a deterministic history space—as proof that the framework is immune to Landauer overhead [4].

However, this framing conflates ontological erasure (the global destruction of information from the state space) with operational erasure. In

a deterministic multiway system, ontological erasure is absent; information remains preserved within the branching network architecture.

Operational erasure refers to the systematic loss of information from a localized subsystem's record. For an observer restricted to a finite causal diamond, filtering unobservable branches is operationally indistinguishably equivalent to true logical erasure. Because the observer's local frame cannot maintain correlations with alternative branchial trajectories, local physical operations carry the exact informational signature of a logically irreversible process.

Globally, the universe remains in a deterministic history space that is information-preserving. Locally, however, the embedded observer's mandatory coarse-graining filter continuously drives the accessible spatial substrate into a highly mixed statistical density matrix. Because the system is entirely closed, this missing path data cannot be exported to an external environment. It is continuously deposited into the local spatial network, manifesting not as kinetic thermal heat, but as disordered, non-planar edge connections and local vertex degree variations.

5.2 The Relational Frame Projection and Ontic Deficits

A defense of asynchronous rewriting architectures relies on global path preservation. Proponents argue that because the overarching multiway graph retains a record of all divergent trajectories, the entropy from history reconciliation is exported to non-local branchial correlations rather than spatial network defects. Under this view, branchial space functions as a digital heat sink, insulating the spatial hypergraph substrate from Landauer dissipation.

This defense fails in a background-independent framework. The branching multiway graph constitutes the objective ontic state space, while a singular timeline is an observer-dependent coordinate projection. To show why this projection cannot shield the continuous classical vacuum, we bound the relationship between the spatial network and the branchial path states. Let the multiway state space be represented by the tensor-product Hilbert space $\mathcal{H}_{\text{multiway}} = \mathcal{H}_{\text{spatial}} \otimes \mathcal{H}_{\text{branchial}}$. The total state is the pure state vector:

$$|\Psi\rangle = \sum_{i=1}^M \sqrt{p_i} |G_i\rangle \otimes |i\rangle$$

where $|G_i\rangle \in \mathcal{H}_{\text{spatial}}$ defines the orthogonal microspatial graph states, $|i\rangle \in \mathcal{H}_{\text{branchial}}$ defines the orthogonal history path states, and $p_i = 1/M$ represents the uniform classical trajectory frequency distribution.

If the process entropy $H_{\text{process}} = \log_2(M)$ is retained as branchial correlations, the spatial substrate and branchial space remain entangled. For a localized observer restricted to a finite causal frame, alternative histories diverging outside the causal diamond are unobservable, requiring a partial trace over these branchial degrees of freedom:

$$\rho_{\text{spatial}} = \text{Tr}_{\text{branchial}}(|\Psi\rangle\langle\Psi|) = \sum_{G \in \Omega_{\text{terminal}}} p(G) |G\rangle\langle G|$$

Evaluating this partial trace yields a highly mixed spatial density matrix ρ_{spatial} , where $p(G)$ is the path-weighted probability of the physical isomorphism class G . The fine-grained information storage capacity of this state is governed by its von Neumann entropy:

$$S(\rho_{\text{spatial}}) = -\text{Tr}(\rho_{\text{spatial}} \log_2 \rho_{\text{spatial}})$$

which corresponds mathematically to the realized classical Shannon macrostate entropy H_{macro} .

This partial trace over branchial histories is structurally identical to frame-dependent horizon projections in semiclassical gravity. While this mixedness is frame-dependent, the relational adjacency matrix provides the exhaustive description of the background-independent substrate. Consequently, this mixing shifts the expectation values of the metric, forcing a physical, non-vanishing stress-energy tensor ($T_{\mu\nu} \neq 0$) rather than a purely epistemic shift in the observer's mind.

6. Information-Theoretic Constraints on Continuum Emergence

Theorem 1. Let $\mathcal{M}$ be the multiway evolution system of a closed spatial hypergraph $\mathcal{H}$ undergoing dimensional reduction from an initially hyper-connected configuration. Let this system be governed by a local rewriting rule set R under the following three axiomatic conditions:

1. **Closed Ontology:** The hypergraph substrate constitutes the complete physical state space, explicitly excluding external environmental reservoirs or non-local informational sinks.
2. **Operational Coarse-Graining:** The embedded observer enforces a singular classical timeline over finite intervals via a partial trace operator ψ over unobservable branchial pathways.
3. **Information Conservation:** The logical entropy generated by the non-injective many-to-one state mappings is conserved within the operational degrees of freedom of the closed system.

Then, the total volume of alternative history pathways M generates an algorithmic process entropy $H_{\text{process}} = \log_2(M)$ that asymptotically outcales the information storage capacity of the realized classical macrostate ensemble Ω_{terminal} , rendering a pure, low-energy continuous continuum vacuum ($T_{\mu\nu} = 0$) thermodynamically forbidden over finite observational timescales.

Proof. I. Process Entropy Execution

Let $\mathcal{M}$ trace the complete multiway history of a discrete system evolving from an initial complete graph K_N . The total volume of valid history pathways generated under the asynchronous execution of the degree-enforcement rule is denoted by M , establishing an algorithmic process entropy of $H_{\text{process}} = \log_2(M)$ bits. As derived in Section 4.4, the strict sequential baseline lower bound for this vertex scale evaluates to $M_{\text{baseline}} \approx 6.3651 \times 10^{20}$ trajectories ($H_{\text{process}} \approx 69.103$ bits). However, under the action of the dynamic degree-enforcement rule, path-branching nonlinearly accelerates. For the empirically audited scale of $N = 8, k = 3$, the exact path volume accumulated by the induction engine reaches $M = 894,757,885,819,817,073,868,800 \approx 8.9476 \times 10^{23}$ paths, evaluating to a realized process entropy of:

$$H_{\text{process}} = \log_2(8.9476 \times 10^{23}) \approx 79.5658 \text{ bits}$$

II. The Macrostate Capacity Boundary

Let $|\Omega_{\text{terminal}}|$ denote the cardinality of unique graph configurations at the terminal horizon. A classical statistical ensemble distributed over these discrete states has a mathematical upper bound on its information storage capacity. The Shannon entropy of a classical probability distribution over this space is capped by the uniform distribution:

$$H_{\text{macro}} = - \sum_{s \in \Omega_{\text{terminal}}} p_s \log_2 p_s \leq \log_2(|\Omega_{\text{terminal}}|)$$

At the simulated scale of $N = 8, k = 3$, the physical classes collapse to $|\Omega_{\text{terminal}}| = 355$. The maximum Shannon entropy of this macrostate is capped by $\log_2(355) \approx 8.4716$ bits.

Computing the realized Shannon entropy from the non-uniform path-weighted distribution over these 355 classes yields $H_{\text{macro}}^{\text{realized}} = 6.6960$ bits. This value remains below the maximizing uniform boundary, consistent with the principle that bounded information capacities scale sub-extensively with system size [7].

III. Reductio ad Absurdum via Information Capacity Clash

Because the ontology is closed, the difference between process entropy and realized macrostate entropy cannot dissipate:

$$\Delta H^{\text{realized}} = H_{\text{process}} - H_{\text{macro}}^{\text{realized}} = 79.5658 - 6.6960 = 72.8698 \text{ bits}$$

This creates a strict information-theoretic contradiction. A classical ensemble over 355 distinct states cannot possess a Shannon entropy greater than 8.4716 bits. Substituting the trajectory frequencies of the simulation identifies a realized entropy gap of exactly 72.8698 bits, which is significantly more pronounced than the analytical baseline lower bound ($\Delta H_{\text{baseline}} \approx 62.4127$ bits) derived from the falling factorial.

IV. Resolution of the Contradiction

The classical vacuum lacks the structural capacity to absorb this Landauer-mandated history erasure. The final state cannot resolve into a classical mixture of unlabeled manifolds; instead, the missing path data must manifest either as a massive proliferation of hidden, labeled gauge variations or as non-local entanglement across branchial space. In either case, this informational overhead forces a structural deviation from a symmetric spatial background, rendering a continuous continuum vacuum ($T_{\mu\nu} = 0$) thermodynamically forbidden over finite observational timescales. $\square$

6.1 Sourcing the Stress-Energy Tensor via Holographic Entropic Corrections

This structural contradiction highlights a tension within discrete gravity. In general relativity, the vacuum is a stable, minimum-energy baseline invariant under coordinate transformations due to continuous symmetries and conservation laws [5]. When the manifold is replaced with a discrete rewriting substrate, the vacuum ceases to be a passive backdrop. Instead, the reliance on asymptotic confluence to establish a unique history forces the substrate to operate as a closed register that reconciles independent evolutionary paths.

Relying on infinite-time limits to stabilize finite-scale properties leaves intermediate states unconstrained. Every branch merger and dimensional transition acts as a logically irreversible modification of topological data, depositing informational payloads directly into the spatial adjacency matrix. Within a closed ontology where space serves as the medium, these variations manifest as alterations to local edge connectivity profiles.

By mapping the relational adjacency matrix of the spatial hypergraph to a tensor network entanglement matrix, individual hyperedges function as localized quantum entanglement bonds. Under this dictionary, the fine-grained von Neumann entropy $S(\rho_{\text{spatial}})$ measures the exact entanglement density of the spatial substrate.

The entropy gap $\Delta H = H_{\text{process}} - H_{\text{macro}}$ dictates that historical path data remains trapped within the branchial-spatial correlations. Exporting this overhead into branchial space prevents local vertex dislocations, but forces the spatial substrate into a mixed statistical state.

According to Jacobson's entanglement equilibrium thermodynamics, a stable, low-energy continuous continuum vacuum satisfying $T_{\mu\nu} = 0$ requires that the entanglement entropy of any localized causal horizon remain at its maximal equilibrium baseline. This baseline obeys the strict area-law constraint:

$$S_{\text{area}} = \frac{A}{4G\hbar}$$

where A is the area of the localized horizon and G is the gravitational constant. To prove why the resulting spatial density matrix ρ_{spatial} cannot represent an empty classical vacuum, the Rindler horizon system is evaluated in the continuous limit. In standard semiclassical gravity, the ideal vacuum state ρ_0 restricted to a localized causal diamond is described by the thermal Rindler density matrix:

$$\rho_0 = \frac{e^{-K}}{\text{Tr}(e^{-K})}$$

where K is the modular Hamiltonian. This ideal state represents the state of maximal entanglement entropy. Reconciling divergent histories forces an operational many-to-one mapping, leaving the spatial hypergraph in a mixed state $\rho_{\text{spatial}} \neq \rho_0$. The distance between the ideal vacuum ρ_0 and the actual mixed spatial state ρ_{spatial} is bounded by the quantum relative entropy, which is strictly non-negative:

$$S_{\text{rel}}(\rho_{\text{spatial}} \parallel \rho_0) = \text{Tr}(\rho_{\text{spatial}} \log \rho_{\text{spatial}}) - \text{Tr}(\rho_{\text{spatial}} \log \rho_0) \geq 0$$

Expanding this relation yields the first law of entanglement entropy:

$$S_{\text{rel}}(\rho_{\text{spatial}} \parallel \rho_0) = \Delta\langle K \rangle - \Delta S_{\text{vN}} \geq 0$$

where $\Delta S_{\text{vN}} = S(\rho_{\text{spatial}}) - S(\rho_0)$ is the change in the von Neumann entropy of the spatial substrate, and $\Delta\langle K \rangle = \text{Tr}(\rho_{\text{spatial}}K) - \text{Tr}(\rho_0K)$ is the change in the expectation value of the modular Hamiltonian. Because the ideal vacuum state ρ_0 maximizes the entanglement entropy, the mixedness of the hypergraph state forces a negative entropy variation:

$$\Delta S_{\text{vN}} < 0$$

To satisfy the non-negativity of relative entropy, there must be a strictly positive modular energy excitation:

$$\Delta\langle K \rangle \geq S_{\text{rel}}(\rho_{\text{spatial}} \parallel \rho_0) + \Delta S_{\text{vN}}$$

Evaluating this modular Hamiltonian excitation reveals its gravitational manifestation. The modular Hamiltonian is direct-proportional to the conformal Killing energy flux across the horizon boundary Σ :

$$\Delta\langle K \rangle = \frac{2\pi}{\hbar} \int_{\Sigma} T_{ab} \xi^a d\Sigma^b$$

where T_{ab} is the stress-energy tensor and ξ^a is the conformal Killing vector field. Because the modular energy excitation is strictly positive ($\Delta\langle K \rangle > 0$), the integral of T_{ab} across the horizon must be non-zero. Therefore, the stress-energy tensor satisfies:

$$T_{ab} \neq 0$$

This establishes that if the pre-geometric spatial substrate relies on tracing over branchial space to achieve confluence, the resulting spatial density matrix ρ_{spatial} is inherently mixed. By Jacobson's equivalence, this informational mixedness manifests gravitationally as a mandatory, non-vanishing energy density, strictly prohibiting a classical, zero-energy vacuum ($T_{\mu\nu} = 0$).

The entropy gap ($\Delta H \approx 62.4127$ bits for the baseline, and $\Delta H \approx 72.8698$ bits in simulation) validates that this overhead cannot be passively accommodated. Rebranding history-erasure as branchial entanglement does not eliminate its physical consequences. Lacking an external environmental sink, the mixedness from branchial-spatial coupling violates the vacuum entanglement equilibrium, forcing a non-vanishing macroscopic baseline ($T_{\mu\nu} \neq 0$) in the continuum limit. A classical vacuum is therefore thermodynamically forbidden over finite timescales.

Sorting the terminal registry weights by path density allows direct analysis of the structural anatomy of the resulting topological defects. At the $N = 8, k = 3$ scale boundary, the path distribution concentrates heavily into non-uniform configurations, with the top 10 dominant macrostates accounting for 31.88% of the total multiway volume. The complete path-frequency distribution of these top 10 classes is plotted in Fig.~3.

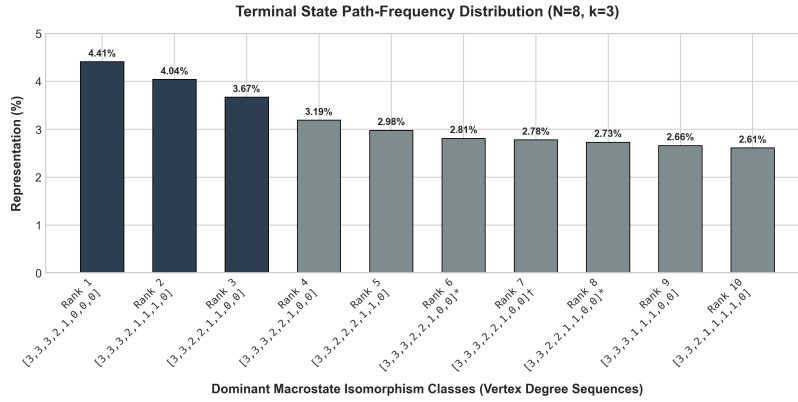


Figure 3: Path-frequency distribution of the top 10 dominant terminal macrostates at the $N = 8, k = 3$ scale boundary, displaying a strong concentration on fragmented degree sequences.

Critically, 100% of this top-tier probability mass is held by “island topologies” exhibiting a pronounced fragmentation pattern: a sparse, connected relational core accompanied by a rapidly proliferating number of completely isolated spectator vertices of degree 0. The single most heavily favored configuration, Rank 1 ([3, 3, 3, 2, 1, 0, 0, 0]), claims 4.41% of the entire global path volume while isolating three out of its eight available nodes entirely, indicating complete physical disconnection from the active relational spacetime fabric.

In Gorard’s framework, particles are localized nonplanar graph defects (subdivisions of K_5 or $K_{3,3}$ as shown in Fig.~4).

However, our path-weighted distribution demonstrates that the vacuum naturally fragments into these nonplanar anomalies and isolated vertices. Matter ($T_{\mu\nu} \neq 0$) does not constitute a rare, stable excitation propagating on a smooth backdrop. Instead, the background metric is systematically altered by topological defects as a consequence of path-merging.

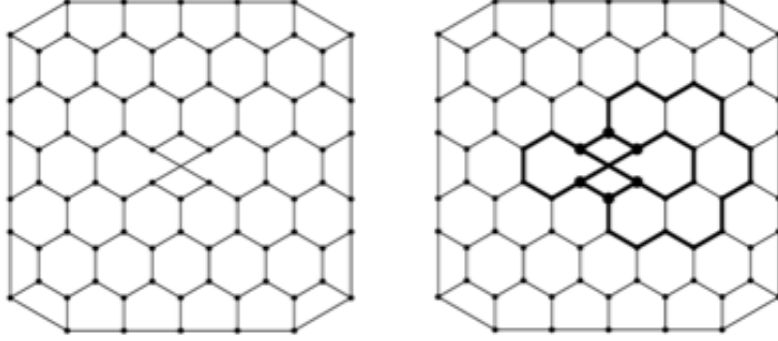


Figure 4: Nonplanar graph defects representing localized particle states, replicated from Ref. [4].

6.2 Explicit Falsifiability Conditions

To ensure this critique is testable, we establish explicit falsifiability conditions. The thesis is falsified if a localized replacement rule set satisfies two criteria simultaneously:

1. Path-Volume Collapse: The rule's multiway evolution concentrates path weights such that the volume of alternative, history-erasing pathways M collapses to 1 as $N \rightarrow \infty$, reducing the process entropy $H_{\text{process}} = \log_2(M)$ to zero.
2. Deterministic Macrostate Concentration: The rule guides the configuration space away from mixing, concentrating 100% of its path weight onto a single isomorphic terminal lattice without observer-dependent coarse-graining.

If a rewriting architecture satisfies these criteria, it achieves finite-time general covariance with zero operational entropy. Otherwise, the combinatorial cost of a sparse universe is paid, and this informational overhead remains trapped within the pre-geometric vacuum metric.

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Simulation Code and Output

The Python script and raw simulation console logs are archived in the source repository: `simulation.py` and `simulation_output.txt`.